

20

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA1220	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Assessment Tool 1 – Application based questions
Weightage:	40%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
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Section / Batch:	CSE(A1)	Date:	16/07/26

Instructions to Students

- Identify the relevant computer architecture concepts, functional units, or performance parameters required to solve the given application-based problem.
- Analyse the interaction between CPU, memory, bus structures, and I/O components in the given scenario.
- Apply appropriate instruction execution cycles, addressing modes, and architectural concepts to determine the correct solution.
- Compute performance metrics such as CPI, clock cycles, CPU execution time, speedup, or benchmarking measures using appropriate formulas and calculations.
- Interpret the calculated results and explain their impact on overall computer system performance.

QUESTIONS

1. A user reports that a computer becomes slower when several applications are opened simultaneously. To analyse the system behaviour, observe the CPU and Memory usage before and after launching multiple applications. Analyse the observed resource utilization to determine how the CPU, memory, and I/O devices interact during program execution and identify the functional unit that is most likely responsible for the performance change.
2. A user wants to compare the performance of two computers to determine which system is expected to execute programs more efficiently. To compare the systems, observe the CPU model, Base Speed, Number of Cores, and Number of Logical Processors for each system. Analyse the observed processor specifications to determine which computer is expected to provide better performance, and justify your conclusion by relating the specifications to CPU execution time and overall processor performance.
3. A programmer wants to understand how an 8085 microprocessor executes instructions during program execution. To examine the instruction execution process, write and execute a program using an 8085 simulator. Observe the changes in the Program Counter (PC), Accumulator (A), and Flag Register after each instruction. Analyse how the fetch, decode, and execute stages contribute to producing the final result.

Code of Conduct

I S. LOKESH (192572151) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S. Lesh

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Q. No	Understanding of Problem Context (20)	Application of Concepts (30)	Problem-Solving Approach (20)	Accuracy of Solution (20)	Clarity (10)	Total (out of 100)
1	4 / 4	3 / 4	4 / 4	4 / 4	3 / 4	
2	4 / 4	3 / 4	3 / 4	3 / 4	4 / 4	
3	4 / 4	3 / 4	4 / 4	3 / 4	3 / 4	
	20 / 20	23 / 30	18 / 20	17 / 20	8 / 10	86 / 100

Course Coordinator

Faculty Incharge

Analysis of CPU, memory and I/O Resource utilization

Procedure

1. open Resource Monitor
2. observe CPU and Memory usage before opening application
3. open multiple application
4. observe CPU, memory, Disk and Network usage
5. Capture Screenshot before and after opening the Application

Observation 1.

- CPU usage was low
- Memory usage was normal
- Disk and Network activity were minimal

Observation 2

- CPU utilization increased
- Memory usage increased
- Disk activity increased while loading application
- Network usage increased due to internet access.

Analysis.

→ The CPU executes instructions, memory stores active Program read/write operation and network handles Communication.

→ Increased CPU and memory are the primary functions units responsible for the observed performance change.

⑧ Comparison of processor performance

Procedure

1. Open Task Manager → Performance
2. Open System Information
3. Record processor specification for both computers
4. Compare the processor details.

Observation

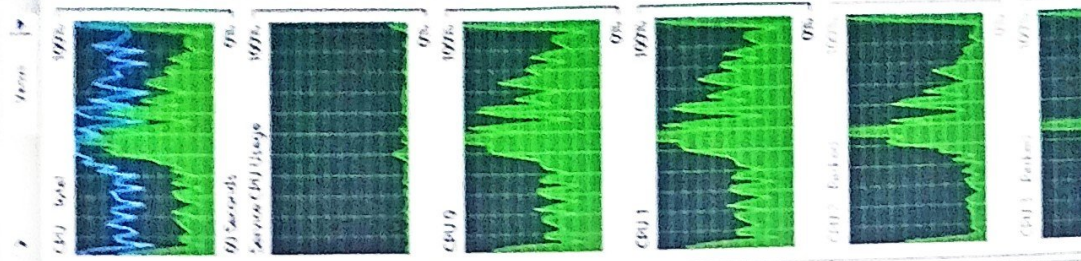
Computer 1 processor specification were recorded and compared with computer 2.

Analysis

→ A processor with higher base speed, more cores, and more logical processor can execute instruction more efficiently and reduce CPU execution time.

Service	Item	Field	Description	Value	Group	Unit	Percentage (%)
Washfield	3336	3336	Washfield Defender Battalion Service	Battalion	Police	1	0.18
Washfield	3336	3336	Washfield Management Instrumentation	Battalion	Police	1	0.18
Washfield	3336	3336	Washfield Category Service	Battalion	Police	0	0.14
Washfield	3336	3336	Washfield Training Service	Battalion	Police	0	0.20
Washfield	3336	3336	Washfield Information	Battalion	Police	1	0.19
Washfield	3336	3336	Washfield Transport	Battalion	Police	0	0.15
Washfield	3336	3336	Washfield Search	Battalion	Police	0	0.12
Washfield	3336	3336	Washfield Clerk	Battalion	Police	0	0.01
Washfield	3336	3336	Washfield Reproductive Service	Battalion	Police	0	0.01
Washfield	3336	3336	Washfield	Battalion	Police	0	0.01

Associated Modules			
Image	File	Module Name	Version
		Full Path	
Socket a process to see results.			



Comparison table.

PARAMETER	COMPUTER 1	COMPUTER 2.
CPU Model	11th Gen Intel Core i3 - 1115G4	13th Gen Intel Core i3 - 1305U
Base speed	2.90 GHz	1.60 GHz
Current speed	2.47 GHz	3.03 GHz
Number of cores	2	5
Logical processors.	4	6

Conclusion

Computer 2 is expected to provide better performance because of its superior specification.

③ 8085 Instruction Execution Analysis.

MVI A, 05H

MVI B, 03H

ADD B

MOV C, A

HLT

Instruction	PC	Accumulator	Flag Register	Observation
MVI A, 05H	changed	05H	No change	Data loaded into Accumulator
MVI B, 03H	changed	05H	No change	Data loaded in Register B
ADD B	changed	08H	updated	Addition performed
MOV C, A	changed	08H	No change	Result copied Register C
HLT	final	08H	No change	Program execution Stopped

Analysis

→ The process fetches the instruction from memory, decodes it, executes the operation, updates the program counter and modifies the Accumulator and Flag register when required.

Result

→ The program executed successfully.

→ The final result (08H) was stored in the Accumulator and the program counter advanced after each instruction.

Processes

Name	Status	55%	81%	5%	0%	Power usage
		CPU	Memory	Disk	Network	
WMI Provider Host		16.9%	15.5 MB	0.1 MB/s	0 Mbps	High
Services and Controller app		13.8%	7.1 MB	0.1 MB/s	0 Mbps	Moderate
> Google Chrome for Testing (8)		6.9%	72.2 MB	2.0 MB/s	0.7 Mbps	Low
> Task Manager		3.1%	80.5 MB	0 MB/s	0 Mbps	Low
System		2.7%	0.1 MB	2.5 MB/s	0.1 Mbps	Low
> Service Host: Windows Manag...		2.7%	22.1 MB	0 MB/s	0 Mbps	Low
> Antimalware Service Executable		1.9%	130.2 MB	0.1 MB/s	0 Mbps	Very low
> EonVPN Gateway Service (32 ...		1.9%	27.4 MB	0.1 MB/s	0 Mbps	Very low
> Google Chrome (15)		1.2%	1,171.0 MB	0.1 MB/s	0.1 Mbps	Very low
Desktop Window Manager		1.2%	41.0 MB	0 MB/s	0 Mbps	Very low
> EonVPN Routing Service (32 bi...		1.2%	15.1 MB	0 MB/s	0 Mbps	Very low
> Windows Shell Experience Ho...		0.8%	32.4 MB	0.1 MB/s	0 Mbps	Very low
> Service Host: Remote Procedu...		0.4%	11.5 MB	0 MB/s	0 Mbps	Very low
Windows Explorer		0%	141.7 MB	0.1 MB/s	0 Mbps	Very low
> Service Host: Network Service		0%	2.4 MB	0 MB/s	0 Mbps	Very low
> Service Host: Application Infor...		0%	1.4 MB	0 MB/s	0 Mbps	Very low
> Microsoft Software Protection ...		0%	1.4 MB	0 MB/s	0 Mbps	Very low
> Service Host: DCOM Server Pr...		0%	11.4 MB	0 MB/s	0 Mbps	Very low
> Service Host: Secure Socket Tu...		0%	0.6 MB	0 MB/s	0 Mbps	Very low
Windows Audio Device Graph ...		0%	11.4 MB	0 MB/s	0 Mbps	Very low
> Service Host: DHCP Client		0%	1.0 MB	0 MB/s	0 Mbps	Very low
> Service Host: SysMain		0%	1.5 MB	0.1 MB/s	0 Mbps	Very low
Microsoft OneDrive Sync Servi...		0%	15.1 MB	0 MB/s	0 Mbps	Very low
> Service Host: Clipboard User S...		0%	4.6 MB	0 MB/s	0 Mbps	Very low
> Microsoft Windows Search Ind...		0%	14.6 MB	0.1 MB/s	0 Mbps	Very low
> Service Host: State Replicator		0%	1.0 MB	0 MB/s	0 Mbps	Very low

© 2014 Core i7-4790K (3.6GHz) / 16GB DDR4

CPU



Usage	Current	Max	Min	Max
69%	3.03 GHz	3.60 GHz	3.60 GHz	3.60 GHz
874	5348	174864	174864	174864
1284858				

Performance



CPU
2.47 GHz



Memory
6.4/7.68 (83%)



Disk 0 (C:)
SSD (NVMe)
1%



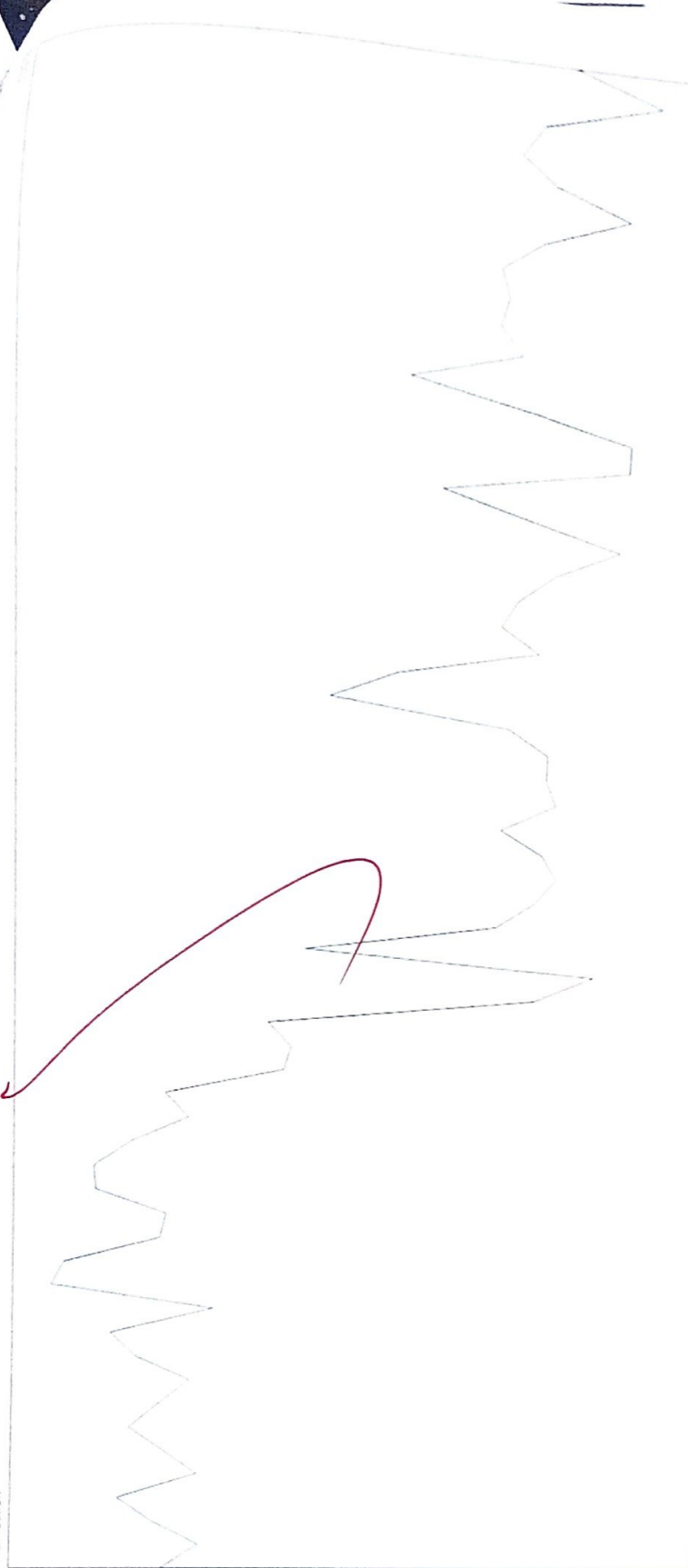
Wi-Fi
Wi-Fi
5.0 R. 0 Kbps



GPU 0
Intel(R) UHD Graphics
0% N/A

CPU

% Utilization



60 seconds

Utilization	Speed	Base speed:
28%	2.47 GHz	2.90 GHz
Processes	Threads	Sockets
252	3000	1
	Handles	Cores
	121248	2
		Logical processors
		4
		Virtualization
		Enabled
		L1 cache
		160 KB
		L2 cache
		2.5 MB
		L3 cache
		6.0 MB

Up time
3:03:49:16

System Summary

- Hardware Resources
- Components
- Software Environment

Item	Value
OS Name	Microsoft Windows 11 Home Single Language
Version	10.0.26200 Build 26200
Other OS Description	Not Available
OS Manufacturer	Microsoft Corporation
System Name	YOGESH
System Manufacturer	Dell Inc.
System Model	Inspiron 15 3530
System Type	x64-based PC
System SKU	0C2F
Processor	13th Gen Intel(R) Core(TM) i5 1305U, 1600 Mhz, 5 Core(s), 6 Logical Process...
BIOS Version/Date	Dell Inc. 1.32.0.23 04/2026
SMBIOS Version	3.6
Embedded Controller Version	1.16
BIOS Mode	UEFI
Baseboard Manufacturer	Dell Inc.
Baseboard Product	0W9CPV
Baseboard Version	A00
Platform Role	Mobile
Secure Boot State	On
PCR7 Configuration	Elevation Required to View
Windows Directory	C:\WINDOWS
System Directory	C:\WINDOWS\system32
Boot Device	\Device\HarddiskVolume1
Locale	United States
Hardware Abstraction Layer	Version = "10.0.26100.1"
User Name	Yogesh\yogesh
Time Zone	India Standard Time
Installed Physical Memory (RAM)	8.00 GB
Total Physical Memory	7.69 GB
Available Physical Memory	1.72 GB
Total Virtual Memory	23.7 GB
Available Virtual Memory	9.80 GB

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